

# SMOTE-WCGAN-GP Hybrid Model for Enhanced Machine Learning Performance

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**Abstract**— This technical paper illustrates the experimental procedures taken to implement a proof of concept. This is an analysis on cause and effect on the hypothesis that the availability of a larger training Dataset will improve the accuracy of Machine learning algorithms in predicting employee attrition. After a comprehensive survey on the available data augmentation tools the proposed solution is to create synthetic data to enlarge an employee dataset without introducing statistical bias by combining Simple Minority Oversampling Technique (SMOTE) and Wasserstein Conditional Generative Adversarial Network with Gradient Penalty (WCGAN-GP). An EEDA was carried out to reveal patterns in data, establish trends and define necessary features. Against its inherent use, SMOTE came in handy as a data augmentation technique when the majority class is also labelled as the minority class. In addition to balancing the dataset it also created new data points. SMOTE ensured that the implementation of WCGAN-GP was stable. After successfully implementing the model a simple hyper parameter tuning was conducted to adjust Latent Dimension, Batch Size, Number of Epochs and Critic training Steps. The table-evaluator module was implemented with the primary objective is to provide a visual assessment of the similarity between the real data (ground truth) and the generated synthetic data by generating multiple plots. The machine learning models were evaluated using the accuracy and the AUC Curve to determine the best performing algorithm either before or after Augmentation. Thereafter Precision, Recall, and F1Score was used to evaluate the effect of data augmentation using the best out of the 4 algorithms. The machine learning algorithms are Support Vector Machines (SVM), Logistic Regression (LR), Random Forest (RF) and Decision Tree Classifier (DTC). The implementation of the model was successful and the synthetic data closely resembled the real. SVM being the best model in both cases it was used to analyse the effects of data augmentation. Based on the classification reports overall accuracy was 87% Before data augmentation and 93% after Augmentation. In all metrics (accuracy, precision, recall, and F1-score), the performance of the algorithm after data augmentation is substantially better than before. This suggests that data augmentation was an effective technique for this dataset.

**Keywords** — Data Augmentation, Attrition, WCGAN-GP, SMOTE, GANs, DTC, SVM, RF, LR

## I. INTRODUCTION

Employee attrition is the gradual reduction in size of a workforce due to employees leaving the company and not being replaced. It is a natural part of any business, but high

levels of attrition can be disruptive and costly [1]. To overcome the negative impacts of employee attrition, the use of predictive machine learning models is useful but is sometimes limited or even biased due to lack of sufficient historical data [2]. After a comprehensive survey we finally proposed, investigations on the potential of combining SMOTE and WCGAN-GP for data augmentation. Since GANs May struggle with data sparsity when there are very few examples of certain combinations of features. SMOTE might help to balance the dataset or even expand it. This will minimize statistical bias and promote generation of realistic data hence improving the effectiveness of employee attrition prediction models [3]. The methodology will focus on drawing strong conclusions about cause-and-effect relationships between variables. The hypothesis here is that the availability of a larger training Dataset will improve the accuracy of Machine learning algorithms in predicting employee attrition and the proposed model will successfully generate Synthetic data without introducing Statistical Bias. The independent variable is the size of the of the training dataset of employee data the depended variables are the model accuracy to be tested after altering the size of the training dataset and the statistical Parameters of data, to be tested after Data augmentation.

## II. METHODOLOGY

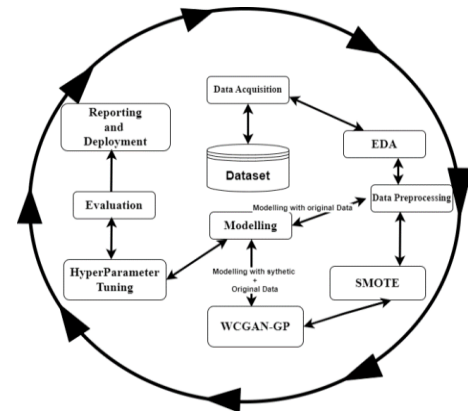


Figure 1 : Proposed Project Methodology (CRISPY-DM)[4]

The methodological working flow of our research study is examined in Figure 1. The type of analysis pipelines to be undertaken fit very well in the Cross Industry Standard Process for Data Mining (CRISP-DM). The methodology CRISPY-DM was chosen as a desire to harness its iterative nature in the flow of processes meaning that processes can be repeated if necessary even when the research is already at the next stage,[4]. The idea of the proposed solution is to create synthetic data to enlarge an employee dataset without introducing statistical bias by combining Simple Minority Oversampling Technique (SMOTE) and Wasserstein Conditional Generative Adversarial Network with Gradient

Penalty (WCGAN-GP). The purpose is to improve the accuracy of employee attrition prediction using machine learning models. The python language is used in this project in a notebook file together with pandas, numpy, sci-py packages and many others that facilitate the manipulation, reproduction, enumeration, plotting, slicing and analysing of datasets and their elements. The machine learning algorithms to be trained before and after data augmentation are Support Vector Machines (SVM), Random Forests (RF), Logistic Regression (LR) and Decision Tree Classifier (DTC).

#### A. Experimental Procedure

The IBM HR Employee Attrition Dataset available at was used for data analytics and model building for the prediction of employee attrition of valuable employees. The dataset was created by IBM data scientists. The dataset contains 35 features. The dataset has 1470 records of employees. The memory usage by data is 402.1 + KB. Useful insights and data patterns were observed from the HR employee attrition dataset through EEDA. The Data Pre-Processing phase aimed to prepare the dataset to be ready for applying ML algorithms in order to obtain the best results. Complex information was presented visually which is easier to understand. By visualizing data, we identified patterns, trends that might be difficult to spot just by looking at raw numbers.

Figure 2 shows a Pie Chart that compares the percentage of positive attrition to negative attrition there is evidence that the dataset is unbalanced since 83.9 % of the target variable is negative (No Attrition) and 16.1% (Attrition). Training our algorithms on an imbalanced dataset may cause bias towards the majority class, misleading accuracy metrics and poor decision boundaries. In addition, it might produce unrealistic synthetic data if used for augmentation, this justifies the need to use SMOTE for dataset Balancing.

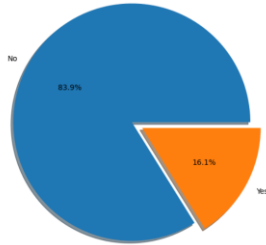


Figure 2 : Attrition Pie Chart Showing Target Variable by Percentage

#### B. Applying SMOTE Initially

Synthetic Minority Oversampling Technique (SMOTE), is a technique originally designed to fix a common problem in machine learning: imbalanced datasets. This happens when one class of interest has far fewer examples than others. SMOTE tackles this by creating new, synthetic data points for the minority class [5]. SMOTE's usefulness goes beyond balancing classes. It can also be used to generate entirely new artificial data. This can be helpful when real data is scarce or expensive to collect [6].

SMOTE was also used on the majority class, treating it like a minority class for a moment. This was a creative way to artificially increase the size of all classes in a dataset, similar to other data augmentation techniques [7]. The mathematical

equation on (1) summarizes the generation of a new instance in the minority class or in the majority in the event that SMOTE is used as an augmentation tool [7].

$$(s = x + \alpha(y - x)) \quad (1)$$

Repeating the process for a predefined number of times oversamples the minority class by creating new synthetic data points based on the existing ones. Against its inherent use, SMOTE came in handy as a data augmentation technique when the majority class is also labelled as the minority class [8]. At least 636 data points were added after performing data augmentation. Figure 3 below shows how the dataset is now balanced with 1590 rows on both yes and no columns of attrition from an original of 954 rows as presented in Fig 4 Before. '0' means No Attrition and '1' means Yes.

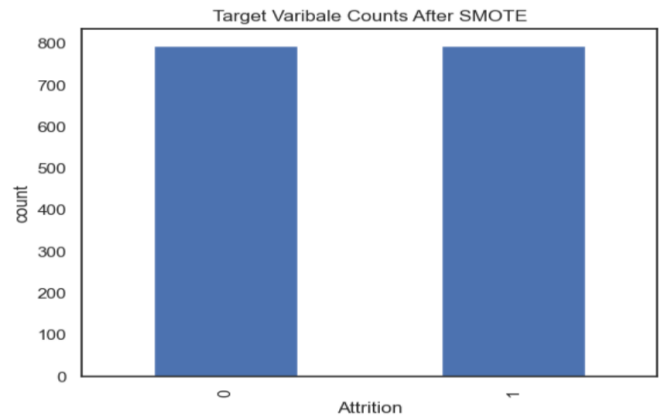


Figure 3: Bar graph Showing the target variable counts after SMOTE

#### C. WCGAN-GP

I) *The Generator(g)* : The Generator's Job is Creating Believable Fakes. The Generator uses Random Noise and Deep Learning Artificial Neural Networks (AANs). (G) starts with a random sample (z) from a special space called the "latent space." The latent space is the feature space of interest that holds all the attributes to be used for creating Synthetic data. The space holds the different relations in data that are used for creating different types of data [9].

$$J_G^{MM-GAN} = \mathbb{E}_{Z \sim p_Z(Z)} [\log(1 - D_{\theta_d}(G_{\theta_g}(z)))] \quad (2)$$

The mathematical formulae above shows how G uses a deep neural network (ANN) to transform this random sample (z) which could be age or salary, derived from the latent space defined by as a set of  $p(z)$  into a new sample that looks like real data. The network learns the best way to do this by adjusting its internal parameters, represented by  $(\theta_g)$  given an input  $(z \sim p_Z(Z))$  that produces the output  $G_{\theta_g}(z)$ . (G)'s ultimate goal is produce output  $G_{\theta_g}(z)$  to trick the discriminator (D) into thinking its creations are real. It wants D to give its generated samples a score close to 1 which shows that samples highly likely to be real due to their closeness to the latent space [9].

The Discriminator ( $d$ ) : For each sample ( $x$ ) generated by ( $G$ ), the discriminator outputs a score,  $D_{\theta_d}(x)$ . This score represents the probability that  $x$  actually came from the real data world defined as ( $P_{data}(x)$ ) and not from the generator ( $p_g$ ). ( $\Theta_g$ ) refers to all the internal parameters that the discriminator has learned during training. Ideally, the discriminator wants to give samples ( $x$ ) coming from ( $P_{data}(x)$ ) a score close to 1 meaning highly likely to be real and a score close to 0 meaning highly likely to be fake. To achieve this, the discriminator uses a special function called a sigmoid function in its final layer [10]. This function squishes any value between 0 and 1, making it perfect for representing probabilities. This function can be represented by equation (3) below as follows.

$$J_D^{MM-GAN} = \mathbb{E}_{x \sim p_{data}(x)} [\log D_{\theta_d}(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D_{\theta_d}(G_{\theta_g}(z)))] \quad (3)$$

II) *GAN Loss* : There's one loss function for the generator and another for the discriminator. This leads us to the concept on the Min-Max Game. The generator aims to minimize its loss, while the discriminator aims to maximize its own loss. This creates a competition that pushes both models to improve [11]. The losses below for D and G are the original formulations as proposed by Good Fellow, called the minimax GAN (MM-GAN). [12]

$$\min_{\theta_g} \max_{\theta_d} \mathbb{E}_{x \sim p_{data}(x)} [\log D_{\theta_d}(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D_{\theta_d}(G_{\theta_g}(z)))] \quad (4)$$

The primary goal is minimising over  $\theta_g$  and maximising over  $\theta_d$ , so typically the training alternate between G, D for  $n$  number of times. The number of cycles are called Epochs, to date there has been debate among researchers on the optimal number of cycles for GANs. The epoch greatly depends on data veracity, sparsity and volume at large. The overall key takeaways postulate that the generator loss encourages the generated data to resemble the real data distribution as much as possible [13].

III) *WCGAN* : WGANs introduce the Wasserstein loss, a metric that provides a more meaningful measure of distance between real and generated data. The discriminator is actually called a critic instead of a discriminator. The critic uses a scoring system instead of a simple classification. The output is no longer a probability but any number which allows distance to be calculated between them hence calculating the Wasserstein Distance. Real samples get high scores, while fakes get low scores. The Wasserstein loss then measures the distance between the probability distributions of synthetic and real samples [13], [14]. Casually known as the Earth Movers Distance (EM) [15] the Wasserstein Distance measures the minimal changes needed to transform  $p_g$  into  $p_{data}$ , where Distance is defined by  $W(p_{data}, p_g)$  and  $p_g$  is a sample “ $x$ ” generated by ( $G$ ) and  $p_{data}$  represents a sample  $y$  from the set of the original data.

The equation below shows a mathematical summary of WCGAN as used in the dataset.

$$W(p_{data}, p_g) = \sup_{\|f\|_L \leq 1} \mathbb{E}_{x \sim p_{data}} [f(x)] - \mathbb{E}_{x \sim p_g} [f(x)] \quad (5)$$

IV) *WCGAN-GP*: A key innovation of WGAN-GP is the gradient penalty. This penalty term is added to the critic's loss function. It helps to enforce a specific constraint on the gradients of the critic, leading to more stable training and preventing issues like vanishing gradients that can plague GAN training [16]. Instead of clipping the weights of the critic directly, WCGAN-GP takes a different approach. It introduces a penalty term called the gradient penalty. This penalty encourages the critic to have gradients with a specific property: a norm of 1 (think of norm as the magnitude). By enforcing this property, WCGAN-GP makes the critic's training more stable and prevents its gradients from exploding [17].

The "Conditional" aspect of WCGAN-GP comes from feeding both the generator and critic with additional information like class labels along with the data itself. This allows the generator to create data samples specific to the provided labels. For instance, if the labels represent different types of clothing (e.g., dress, shirt, pants), the generator can learn to produce images of these clothing items based on the corresponding labels it receives. Generally, WCGAN-GP is considered an improvement over the original WGAN due to its effectiveness in addressing mode collapse and training stability. Like other GANs, WCGAN-GP requires careful hyperparameter tuning to achieve optimal results. The Conditional GAN (CGAN) which incorporates “an extra” class of labels by both the Discriminator and the Generator Network [9]. For example, to generate synthetic data points for an age column we will have to specify a threshold based on the average life expectancy of an average human and the legal retirement age. In this case this will be our Condition e.g. 16 – 65, will make sure the variations of synthetic data points fall within that range. However, arguments suggest that the techniques' performance cannot be generalised over different column data types since it has only proven to be useful in numerical Data types [18]. In the employee dataset WCGAN-GP was implemented on the Smote Generated Training Data plus the original data, the size of the Dataset was now enormous.

#### Parameter Tuning

The latent dimension refers to the size of the noise vector used as input to the generator network. The generator Network will then take the noise vector as input. It uses its internal architecture to map it to a realistic output like an image or an attribute of a feature in tabular data. For this research latent dimension of 64 as a way to save the limited computational power.

At the same time without affecting the quality and diversity of generated data and retaining control over the generator. The batch size is the number of data samples processed together during a single training iteration during training. The batch size determines how many of these samples are grouped together for each training step. A batch size of 256 was used so as to achieve a smoother gradient, potentially resulting in faster convergence and may improve efficiency. Epoch essentially represents one cycle where the generator and critic networks are trained on all available data samples. As the WCGAN-GP progresses through epochs, the generator learns to create more realistic outputs, while the critic hones its ability to distinguish real data from generated samples. The research used 15 epochs to determine how much the model is exposed to the training data and ensure model effective learning. The total number of times the critic network is trained on a batch of data within a single epoch Epoch is known as the critic steps. The number of critic training steps per epoch determines how often the critic is updated on new batches of data within a single epoch if not defined by user the system has default values that it assumes usually the same as the generator steps. The research used 5 critic steps to ensure the critic learn effectively without overfitting. [19].

#### D. Model FlowChat

Figure 7 Below shows a Model Design Flow Chart that describes the steps taken after acquiring the dataset from pre-processing to target variable balancing through SMOTE then Augmentation by WCGAN and finally algorithm training and evaluation to ascertain the best of the machine learning model for deployment together with the augmentation Model.

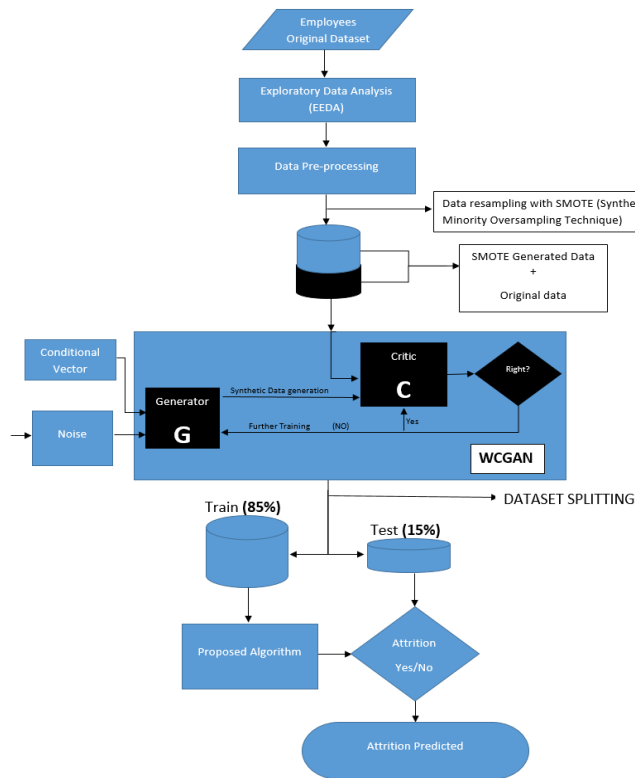


Figure 4: Model Flow Char

### III. RESULTS AND ANALYSIS

#### F. SMOTE Results

The first step is to acknowledge the successful application of SMOTE to our dataset. Not only was the imbalance problem solved because SMOTE acted as a data augmentation method as shown in the bar graph presented before. Figure 5 below shows 2 pie charts comparing the target variable counts by percentage (%) before and after SMOTE balancing. The pie charts clearly show that the dataset is now balanced and is ready for WCGAN-GP.

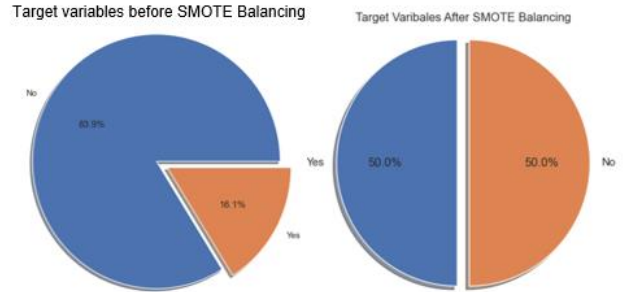


Figure 5 : Target Variable Counts by % Before and After SMOTE

#### G. WCGAN-GP Results

WCGAN-GP was successfully implemented and next was the generation of synthetic data points. We will consider our synthetic data as an effective representation of the real data only if it aligns with the statistical properties of the real data used when creating it. To evaluate the closeness of the synthetic data to the real data the table-evaluator module in python is then used. The research implemented the table-evaluator function in Python's table evaluator module, specifically designed to create multiple plots that visually compare real and synthetic tabular datasets. This function's primary objective is to provide a visual assessment of the similarity between the real data (ground truth) and the generated synthetic data. By generating multiple plots, it allows for analysis of different aspects of the data distributions and identify potential discrepancies between the two datasets.

#### H. Synthetic Data Evaluation

Absolute Log Mean and Standard Deviation provides a measure of the central tendency of a distribution, but on a logarithmic scale. It was particularly useful for comparing distributions that have skewed characteristics or cover a wide range of values. Below are plots that compares the Absolute Log mean and STDs for all the numeric features. The synthetic plots align with that of real data which suggests the synthetic data replicates the central tendency of the real data.

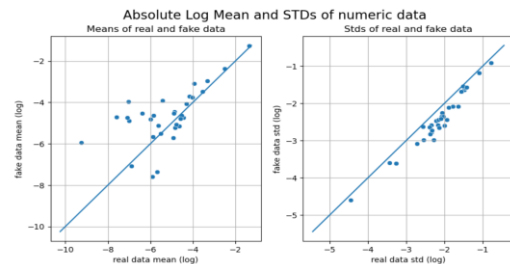


Figure 6 : Absolute Mean and STDs of Numeric Data

The use of scatter plots to plot cumulative sums of data in numerical columns reveals trends in the data. Plotting the real

and synthetic data sums on the same graph allows for visual comparisons of the potential discrepancies. Figure 7 below shows how the synthetic data cumulative sums align well with those of real data, showing that there are significant similarities in the original and the generated training datasets.

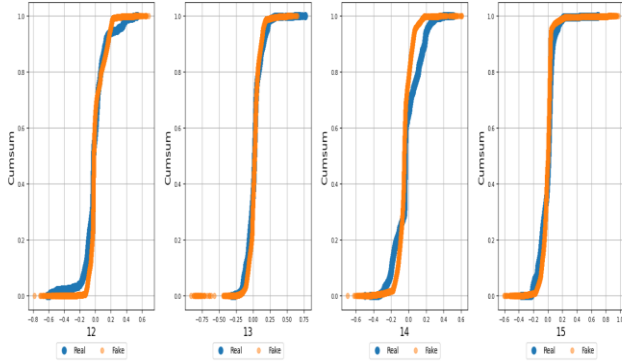


Figure 7 : Cumulative Sums Plot

To support the above line graphs, we proceeded to plot Distribution Bar Plots for each feature. The main goal was to ensure resemblance and minimal divergence. A comparison of the feature distributions can be found in Figure 13, The grey coloured areas show where the synthetic data and real data distributions intersected and this is great evidence that there was minimal divergence from the original data distribution.

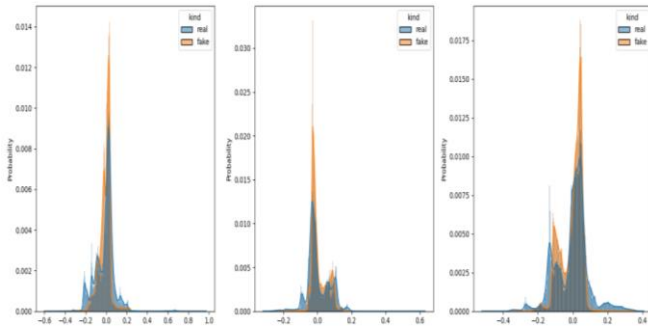


Figure 8 : Distribution Per Feature

### I. Classification/Prediction Evaluation

The performance of the Original and Enhanced Dataset on machine learning models was evaluated using the accuracy and the AUC Curve to determine the best performing algorithm either before or after Augmentation. Thereafter Precision, Recall, and F1Score was used to evaluate the effect of data augmentation using the best out of the 4 algorithms, Support Vector Machines (SVM), Logistic Regression (LR), Random Forest (RF) and Decision Tree Classifier (DTC).

Table 1 : Accuracy, Precision, Recall and F1-Score Formula

| Metric    | Formulae                                                                                            |
|-----------|-----------------------------------------------------------------------------------------------------|
| Accuracy  | $\frac{TP + TN}{TP + TN + FP + FN}$                                                                 |
| Precision | $\frac{TP}{TP + FP}$                                                                                |
| Recall    | $\frac{TP}{TP + FN}$                                                                                |
| F1-Score  | $2 \times \left( \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \right)$ |

Table 1 above shows the evaluation metrics used on this dataset and the formulas used to calculate them based on the confusion matrix values. Accuracy, Precision, Recall, and F1-Score are all important metrics used to evaluate the performance of classification models in machine learning. In addition the purpose of plotting the Area Under the ROC Curve (AUC) when evaluating machine learning algorithms, particularly for binary classification problems, is to assess the model's ability to distinguish between positive and negative classes across various classification thresholds[20]. AUC provides a single, concise metric (between 0 and 1) that captures the model's overall performance in separating the classes.

### I) Classification Results Before Data Augmentation :

Table 2 : Ranking By Accuracy score

|   | Algorithm                     | Accuracy Score(%) |
|---|-------------------------------|-------------------|
| 0 | Support Vector Machines[SVM]  | 87.00             |
| 1 | Logistic Regression[LR]       | 86.58             |
| 2 | Random Forest Classifier[RFC] | 85.74             |
| 3 | Decision Tree Classifier[DTC] | 81.13             |

Table 3 : Raking By AUC

|   | Algorithm                     | AUC  |
|---|-------------------------------|------|
| 0 | Support Vector Machines[SVM]  | 0.72 |
| 1 | Logistic Regression[LR]       | 0.71 |
| 2 | Decision Tree Classifier[DTC] | 0.68 |
| 3 | Random Forest Classifier[RFC] | 0.58 |

Table 3 and 4 above shows the algorithms ranking by Accuracy and AUC before the augmentation of Data. In both Cases Support Vector Machines (SVM) has the best performance with an accuracy of 87% and an AUC of 0.72 followed by Logistic Regression(LR), then Random Forest Classifier (RFC) and lastly Decision Tree Classifier (DTC) when using the metrics accuracy. Random Forest Classifier (RFC) and Decision Tree Classifier (DTC) exchange positions when considering the metric AUC. This is why it is important to incorporate other metrics of measuring algorithm performance to complement the commonly used such as Accuracy and ROC AUC. Figure 18 and 19 visually depicts the results of the Algorithms in forms of bar charts clearly revealing the performance differences

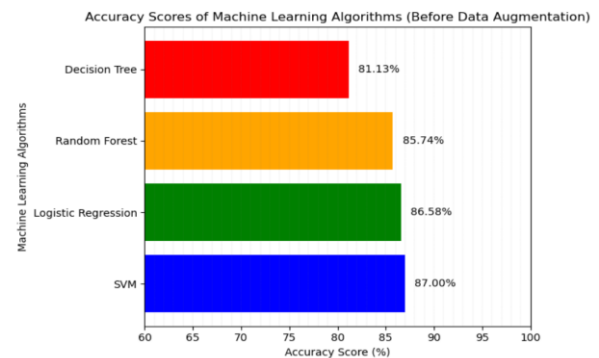


Figure 9 : Accuracy Bar Chart



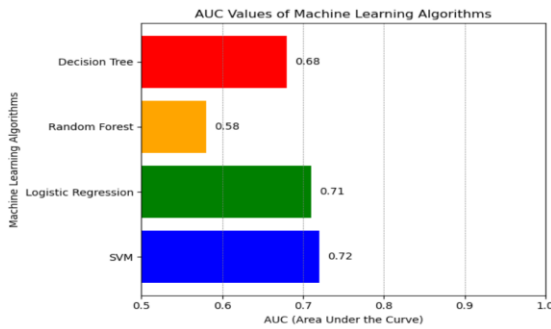


Figure 10 : AUC Bar Chart

## II) Classification Results After Data Augmentation:

Table 4 : : Accuracy Ranking (After Data Augmentation)

|   | Algorithm                     | Accuracy Score(%) |
|---|-------------------------------|-------------------|
| 0 | Support Vector Machines[SVM]  | 93.52             |
| 1 | Random Forest Classifier[RFC] | 88.99             |
| 2 | Decision Tree Classifier[DTC] | 87.61             |
| 3 | Logistic Regression[LR]       | 81.95             |

Table 5 : AUC Ranking (After Data Augmentation)

|   | Algorithm                     | AUC  |
|---|-------------------------------|------|
| 0 | Support Vector Machines[SVM]  | 0.93 |
| 1 | Logistic Regression[LR]       | 0.89 |
| 2 | Random Forest Classifier[RFC] | 0.88 |
| 3 | Decision Tree Classifier[DTC] | 0.82 |

Table 5 and 6 above shows the algorithms ranking by Accuracy and AUC after the augmentation of Data. In both Cases Support Vector Machines (SVM) has the best performance with an accuracy of 93% and an AUC of 0.93. it is evident that the SVM has been the best algorithm both before and after Data Augmentation. For this reason, we will use SVM to evaluate the effect of data augmentation on the performance of machine learning algorithms using the classification Report. Figure 11 Below visually depicts the comparison of Accuracy before and after Data Augmentation. The results show that Support Vector Machines (SVM), Random Forest Classifier (RFC) and Decision Tree Classifier (DTC) improved their accuracy after the training data was Augmented. SVM rose to 93.52%, RFC to 88.99% and DTC to 87.61%.

## III) Classification Results on Real vs Synthetic Data

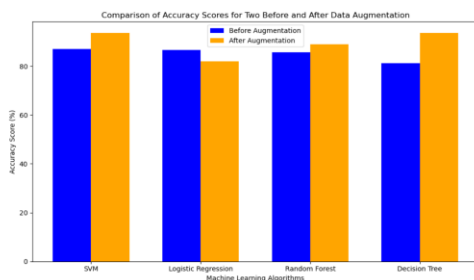


Figure 11 : Accuracy comparison Before and After Augmentation

However Logistic Regression dropped its accuracy to 81.95%. The reason might have been that Logistic regression is a process of modelling the probability of a discrete outcome given a discrete input variable. If the chosen augmentations don't reflect real-world variations or preserve the discrete nature of features, they can actually introduce noise and make the logistic regression model perform worse due to the increase in the number of qualitative features.

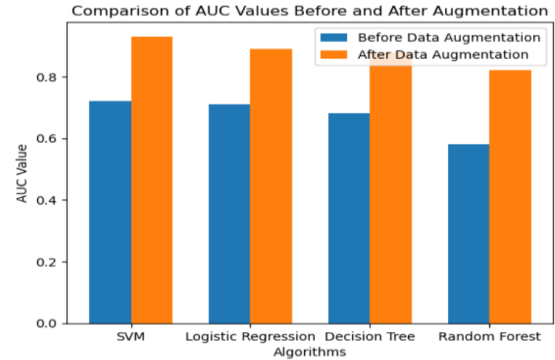


Figure 12 : Comparison of AUC Before And After Augmentation

| SVM Classification Report (Before Data Augmentation) |           |        |          |         | Classification Report (After Data Augmentation) |           |        |          |         |
|------------------------------------------------------|-----------|--------|----------|---------|-------------------------------------------------|-----------|--------|----------|---------|
|                                                      | precision | recall | f1-score | support |                                                 | precision | recall | f1-score | support |
| 0                                                    | 0.90      | 0.95   | 0.92     | 795     | 0                                               | 1.00      | 0.87   | 0.93     | 795     |
| 1                                                    | 0.65      | 0.40   | 0.55     | 159     | 1                                               | 0.88      | 1.00   | 0.94     | 795     |
| accuracy                                             |           |        | 0.87     | 954     | accuracy                                        |           |        | 0.93     | 1590    |
| macro avg                                            | 0.77      | 0.72   | 0.74     | 954     | macro avg                                       | 0.94      | 0.93   | 0.93     | 1590    |
| weighted avg                                         | 0.86      | 0.87   | 0.86     | 954     | weighted avg                                    | 0.94      | 0.93   | 0.93     | 1590    |

Figure 13 : SVM Classification Reports

Fig 24 shows the classification reports one labelled "SVM Classification Report (Before Data Augmentation)" and "Classification Report (After Data Augmentation)". Based on the classification reports overall accuracy was 87% Before data augmentation and 93% after Augmentation. In all metrics (accuracy, precision, recall, and F1-score), the performance of the algorithm after data augmentation is substantially better than before. This suggests that data augmentation was an effective technique for this dataset. From the results it shows that the dataset may have adopted some imbalance. This is because the precision for class 0 is higher for class 1 in both reports, while the recall for class 0 is lower than for class 1 in both reports.

## IV. CONCLUSION AND RECOMENDATIONS

This research aimed to create artificial data that closely resembled an original employee tabular dataset. We achieved this by developing a hybrid model. This method involved training WCGAN-GP a neural network model on a balanced version of the employee data created using a technique called SMOTE. By comparing different similarity metrics, they showed that the synthetic data generated by this approach closely resembled the original dataset. Traditionally, unbalanced tabular datasets haven't been augmented because they're seen as lacking complexity. However, this approach

successfully generated synthetic data using WCGAN-GP. The key was starting with an SMOTE augmentation step. This helped explore the data's full distribution more thoroughly, leading to a more stable training process for WCGAN-GP. While SMOTE is common for addressing class imbalance in tabular data, combining it with GANs for augmentation has been rare.

Past research used simpler GANs with SMOTE to improve minority class data, not for overall data augmentation. This is likely due to concerns about overfitting and the potential for SMOTE to introduce unrealistic data points, especially if applied to all classes. However, our approach pushed the boundaries of SMOTE overcame these limitations. Instead of just oversampling the minority class, we applied it to the majority class as well to enlarge the training data for WCGAN-GP. To prevent overfitting due to SMOTE augmentation, we performed basic hyperparameter tuning of the WCGAN-GP model. We adjusted training epochs and latent dimension to capture the data's complexity efficiently. While we didn't optimize the hyperparameters for this specific dataset, they might need adjustment for different data structures.

However, this approach can serve as a valuable framework for future studies.. By carefully comparing the generated data to the original set, we demonstrated that our method using SMOTE with WCGAN- GP produced accurate representations of the underlying data distribution. Data augmentation generally boosted machine learning performance, as measured by AUROC, precision, recall, and accuracy scores across most models. However, logistic regression performed slightly better on the original data. This suggests that for this specific model, the augmentation process may have introduced noise or blurred the discrete nature of the original data, making it harder for logistic regression to learn effectively. Overall, this study highlights the potential of data augmentation, especially for fields with limited data. The hybrid approach used here not only increased data quantity but also aimed to preserve the quality and underlying patterns of the original data. Additionally, it facilitated the creation of more diverse data, which is crucial for robust model training techniques might better retain the original data's characteristics, leading to even better WCGAN-GP training.

## J. Recommendations and Future Work

It is important to remember that this was a proof-of-concept. Techniques like feature extraction, extensive pre-processing, and longer training times could significantly improve machine learning model performance. This approach could be further improved by exploring different SMOTE variations like SMOTE-NC, borderline-SMOTE, and ADASYN to make the generated dataset more compatible with WCGAN-GP. Lastly the evaluation of the synthetic data against the original dataset can be done better by the use of other packages that supports comparison on categorical features and qualitative features accurately unlike Table Evaluator in python which mostly is more accurate on numerical values.

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